CoRoT-2b

R-0113

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_170503` · created 2026-07-16T04:20:50+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457875.1736 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.065
Transit depth (Rp/Rs)^2	4.99 +/- 2.9 [%]
Orbital Inclination (inc)	84.56 +/- 2.1
Airmass coefficient 1 (a1)	1.71 +/- 0.25
Airmass coefficient 2 (a2)	-0.216 +/- 0.042
Scatter in the residuals of the lightcurve fit is	14.24 %
Best Comparison Star	#3 - [555, 399]
Optimal Aperture	17.53
Optimal Annulus	29.21
Transit Duration (day)	0.0165 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.065 vs 0.1667 (deviation 0.87σ)
Duration fit vs published	0.0165 d vs 0.095 d (deviation 10.61σ)
Mid-time O–C	-2506.5 min
Depth SNR	3.4
Residual scatter	14.24 %

Light curve

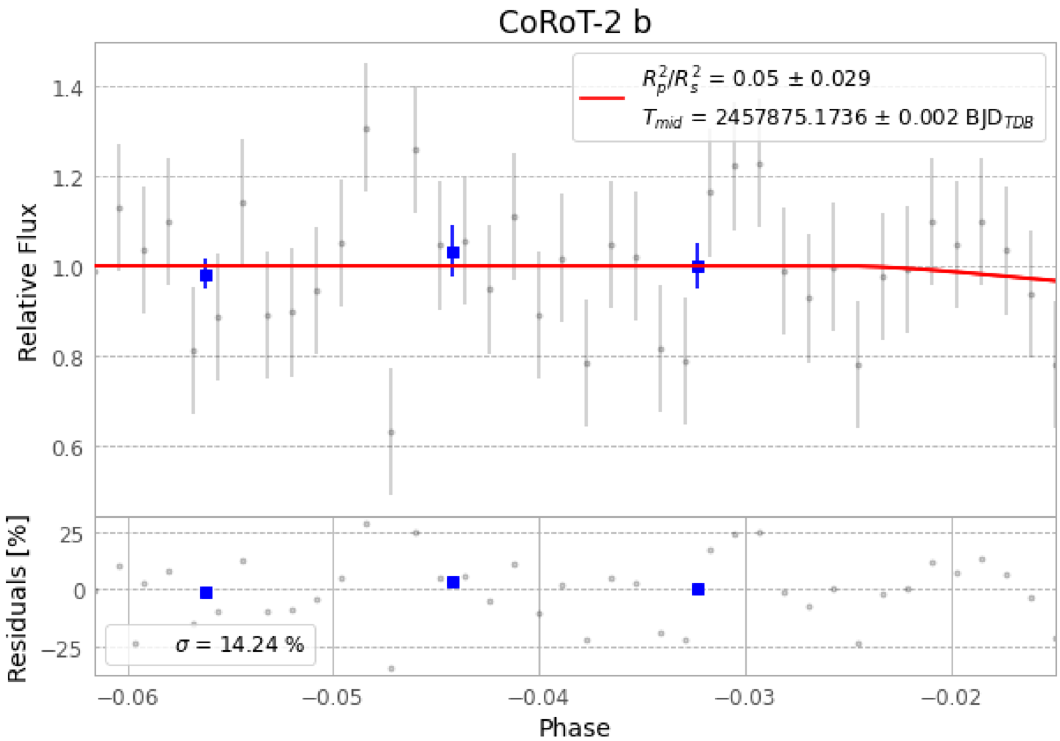

Figure 1 — FinalLightCurve_CoRoT-2 b_2017-05-03.png (SHA-256 15f93d8901ee2764…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [464, 411], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 db92fbdadaaf2717…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7`

Data provenance

87 FITS frames (57 MB), `CoRoT-2170503072112.FITS` → `CoRoT-2170503113912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-170503.log (b80f299ac815…), FinalLightCurve_CoRoT-2 b_2017-05-03.png (15f93d8901ee…), FinalLightCurve_CoRoT-2 b_2017-05-03.csv (86ef831c5e40…)

Compute resources

Wall time 205.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

